Direct signs expose the signed-cache tradeoff

Quadratic-twist study, PR2381

25 October 2026; direct-sign-20261025

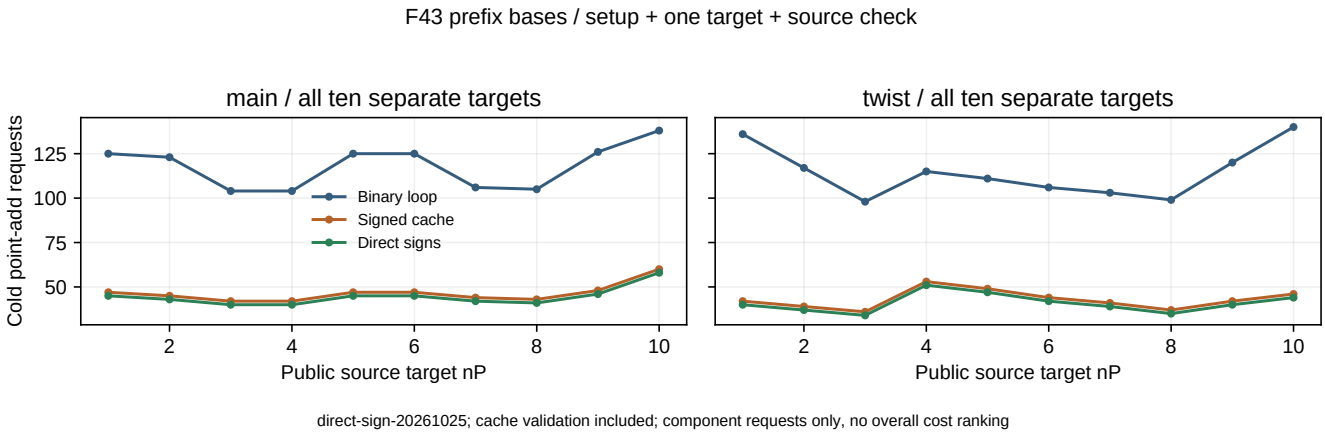
Executed comparison. Native C++ and independent Rust execute 120 setups and 1,200 separate nonidentity main-source recoveries,40/400 per arm: binary, direct uncached and setup-validated cached signs. All ten F43 size-two policies/four roles remain; all 1,200 scalars verify. Both old arms match frozen records. Every root, ordered lift, row/rank, full certificate, degree/witness, promotion, base log, failed shift and scalar is equal. New hashes bind rejected operand/sum tests too. A separate batch-RREF/witness audit checks 100 certificates.

A stronger reference changes the interpretation

Direct uses P or one negation per operand without the binary scalar loop. Cached setup still charges its entries, curve/sign checks and two inverse validation additions. All signs in each query are tested even after a success.

Prefix policy, separate $Q = P$; cold requests	Binary	Direct	Cached
Main point-add requests	125	45	47
Twist point-add requests	136	40	42
Main negations	20	20	2
Twist negations	24	24	2
Main scalar-loop iterations	54	14	14
Twist scalar-loop iterations	56	8	8
Main signed-operand reads	40	40	40
Twist signed-operand reads	48	48	48

All 400 direct cold records use two fewer additions than cached. All 400 cached records use fewer negations. Direct removes $4L$ additions and$2L$ scalar-loop iterations relative to binary for L tested signs, retaining its negations. Cache replaces those negations by two at setup, at the cost of validation, storage/reads. Neither arm dominates every request component.

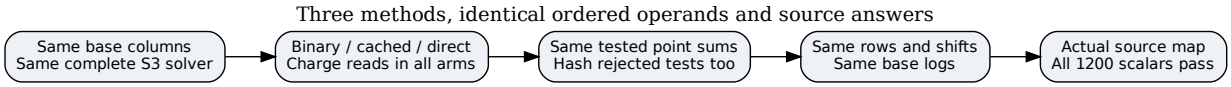


The old binary loop includes identity additions/unused doublings, which compilers may optimize. These 38 differently weighted/nested requests are not retired instructions or runtime. Cache checks/storage/reads, field tables, hashing/trace work, allocation/sorting/traversal, exponentiation and audit remain unpriced. No complete-work ranking, speedup, GAE, S, rho, intrinsic regularity or fastest-external claim is admitted. The earlier component reduction is valid against binary; it does not establish the value of caching against direct signs.

Exact operands and existing proof bridge

Every query binds the complete ordered tested tuples $(i, j, s_i, s_j, A, B, A+B, \text{accepted})$, including rejected signs. Rust independently reconstructs these hashes, all 38 counters and the complete mathematical outputs. Two new common counters, sign-operand reads and sign dispatches, equal $2L$ in all arms. The older cache-only read counter is nested and must not be counted again as equal-cost work. Hashes supply no solver answer.

Direct is the existing Lean `SignedCache.signedPoint` branch. `signed_eq_zsmul`, `ordered_lifts_eq`, validated cache and actual recovery theorems already prove the required group/list implications. The actual cache, base logs and source annihilator are checked. No duplicate theorem/new polynomial identity is introduced. The unmodified scoped gate again passes 191 public/249 audited declarations, zero failing/extra trusted; only propext, Classical.choice and Quot.sound. All 16 identity records remain; `IDC1h202bb532af10c977` is rechecked with unchanged source/statement and false identity/source controls. Native request execution is independently replayed, not universally formalized.



Actual source DLP and all-policy results

Main `icv1-fp6-t0-74779950` is$E : y^2 = x^3 + x$, $P = (31, 18)$; twist `icv1-fp6-t0-c6a8a47c` is$T : Y^2 = X^3 + 4X$, $U = (24, 26)$ over F43, selected order11. The actual additive/injective map$(x, y) \mapsto (41x, 11y)$ binds source recovery. This exceptional pair is base-field-isomorphic; U has no EC1 registration. Recovery receives the point only, with labels outside it.

Summed hypothetical cold main/twist additions are 11750/11990 (binary),4398/4638 (cached),4198/4438 (direct). Negations are 1888/1888,200/200,1888/1888 respectively. These sum 100 separate one-target records per role, not batch throughput or independent curves. Direct/cached twist-addition lower/equal/higher counts remain 43/1/56. Earlier first-query rank0/2, operational degree3/2 and equal homogeneous cutoff2 remain. Both full setups reach coefficient rank2.

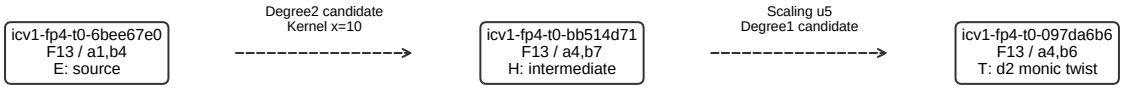
Seven rehashed controls reject without receipt: false direct trace/demand, wrong on-curve cached sign point, missing arm/failed query, false scalar/source. The fresh replay and 22 backend tests pass; full library and metadata checks are recorded in validation. Profile remains unavailable through missing gawk, Conductor exits127, and no paid resource/upload occurred.

Next actual transfer/yield lead

The next proposed route uses already registered F13 models `icv1-fp4-t0-6bee67e0` (a1,b4), `icv1-fp4-t0-bb514d71` (a4,b7) and `icv1-fp4-t0-097da6b6` (a4,b6). A standard degree-two coordinate candidate, including the final scaling to the d2 monic twist, is

$$x' = 12(x + 2/(x-10)), \qquad y' = 8y(1 - 2/(x-10)^2).$$

PROPOSED: exact models registered; map and subgroup checks pending



This is a proposed construction, not an executed reduction here. Check infinity/kernel/poles, actual equations, r7 scalar transport/injectivity and non-isomorphism before comparing all size-two bases and useful S3 rows/degrees. Matched-base conservation and all losses remain. No registry/catalog change or upload is needed for these existing models. The cubic j0 option is deferred, not refuted. General geometry, stationary yield, calibrated cost, shared coordination, review and CI/integration remain open. Full sources, proof links, commands and receipts are in the accompanying README.